

# Linear Simultaneous Equations

LET – Mathematics and Statistics

## Two Unknowns

Solve the following systems of equations

1.

$$5x + 2y = 24$$

$$y = x + 5$$

2.

$$8x - y = 34$$

$$3x + y = 10$$

3.

$$10x + 3y = 25$$

$$2y - 6x = 8$$

4.

$$P = \frac{71}{2} - 5Q$$

$$7P - 3Q = \frac{193}{2}$$

## Three Unknowns

5.

$$y = 50 - 5x$$

$$z = 4x + 7$$

$$y = x + 38$$

6.

$$x - y + 4z = 2$$

$$2x + y + z = 15$$

$$5x + z = 21$$

7.

$$3x + 2y - 3z = 2$$

$$2x + y + z = 10$$

$$x + 3y + 2z = 12$$